Mod extension PCB/prototype board

VOLTS1 and VOLTS2 are ranging:
– ≈1.5V–≈18V (for 18V supply)
– ≈0.8V–≈9V (for 9V supply)

With STARVE switch engaged:
– ≈1.5V–≈9V (for 18V supply)
– ≈0.8V–≈4.5V (for 9V supply)

Center negative
9V–18V
J3
RLED 10k
D6 LED
D5 1N5819
C17 100n film
C18 47u electrolytic 25V

Will work at 9V, but headroom will be reduced.
Clean blend may clip on strong transients.
18V recommended.

R25 47
R28 47
C19 100n film
C20 220u electrolytic 25V
C21 100n film
C30 220u electrolytic 25V
R45 47k
R46 47k
C44 100n film
C37 47u electrolytic 25V
C26 100n ceramic
C28 100n ceramic
C29 100n ceramic
U1C TL072
U2C TL072
U3C TL072
R31 10k
Q2 J113
R40 47
R44 10k
VOLTS2 A1M
R33 1k
C38 47u 25V electrolytic
C23 100u electrolytic 25V
STARVE–2
R32 47
C39 100u electrolytic 25V
R36 47
C40 100u electrolytic 25V
R26 10k
Q1 J113
R29 47
R30 10k
VOLTS1 A1M
R27 1k
C32 47u 25V electrolytic
C31 100u electrolytic 25V
STARVE–1
R54 1k
C46 100n film
C47 1n ceramic
R53 100k
C56 1u electrolytic/25V
R56 10k
R55 47k
C53 1u electrolytic/25V
R57 100R
C50 10n ceramic
A10k CLEAN
R34 100R
C25 100n film
R35 2M2
C24 22p ceramic
R58 2M2
C52 220n film
R39 47k
R37 47k
C51 1u electrolytic/25V
R41 100R
C49 220n film
R50 100k
C55 220n film
R49 100k
R48 100k
C54 1u electrolytic/25V
R51 100R
C45 10n ceramic
R52 100k
TAME ≈ 21.95 kHz → 495 Hz
TIGHT ≈ 49 Hz → 258 Hz
C43 68n film
TIGHT C100k
R47 10k
R38 2k2
TAME A100k
C41 100n film
C27 3n3 film
R42 2M2
C42 100n film
R43 470k

Main/stock Muff PCB

No connection to power supply rail
C14 100u electrolytic

C14, C5 and C8 are not needed
(you don't need to solder them on)

Don't solder D1 & D2
(or cut them off).
REMOVED
D1 1n4148
D2 1n4148
D3 1n4148
D4 1n4148

SPLIT
R4 22k
C2 560p
R3 390k
C3 100n
C5 47n
R10 12k
C6 560p
R9 390k
C7 100n
C8 47n
R15 22k
C9 560p
R14 390k
R12 8k2

Do not solder top ends of
R4, R10, R15, R20, R21.
Instead wire them to the
extension board as shown.

C10 3n9
R18 39k
R17 39k
C11 10n
Tone B100k

Do not solder the pots onto PCB,
use wires. Use GAIN/SUSTAIN pot
pin 3 PCB hole to add a jumper
wire to pin 2 hole of the TONE
pot. Thus sending the signal
from T1 straight to T4.

Do not solder the pots
onto PCB, use wires. Use
PCB TONE pot hole 2 to
add a jumper to receive
the signal from T1.
While TONE pot pin 2 is
connected to the VOLUME
pot pin 3 (bypassing PCB).

Do not solder VOLUME pot onto PCB at all.
Instead Use PCB VOLUME pin holes to add
the RC filter. Solder R24 across pin 3 and
2 holes, and solder C16 across pin 2 and 1
PCB holes. And solder a jumper wire to PCB
OUT (which is shorted with VOLUME pot PCB
pin 2 hole). And that jumper wire goes to
GAIN pot pin 3 (bypassing PCB). Thus your
T1 goes to T4 and then goes into the GAIN
pot before T2.

You don't need
to cut anything
here, just make
sure VOLUME pot
pin 2 goes to
extension mod
PCB input, and
extension mod
PCB output goes
to the output
jack.

J1 In
J2 Out
R1 33k
C1 100n
R23 1M
C15 100p ceramic ADDED
R2 82k
R5 820R
C33 10u ADDED
T1 2n2369a hFE 40-60
A100k Gain
R6 1k
C4 100n
R7 8k2
R8 82k
R11 150R
C34 10u ADDED
T2 2n2369a hFE 40-60
R13 82k
R16 820R
C35 10u ADDED
T3 2n2369a hFE 40-60
C12 100n
R20 390k
R21 12k
C22 4p7 ADDED
R19 100k
R22 2k7
C36 10u ADDED
T4 2n2369a hFE 40-60
C13 100n
R24 100R
C16 100p ceramic ADDED
Volume A100k

N.B.
Musikding.de parts list has an error
R1 is supposed to be 33k, not 3k3.

This mod:

1. Removes clipping diodes completely, letting the transistors clip instead, making classic Muff circuit loud and wild, more dynamic, like a real fuzz pedal.
2. Re–routes T4 gain stage to go after T1, giving the clipping stages more pre–gain as they require much more to really clip, since there are no diodes, only transistors clipping when they can amplify no more. And you don't need the final VOLUME boost at all, since the clipping stages without the clipping diodes go LOUD! Well, except at starving/gated voltages not as much.
3. VOLUME knob goes right after the TONE knob after the clipping stages, and then the signal goes to a unity–gain buffer.
4. Adds buffered rat–style tone control (TAME knob, low–pass filter) on top of existing Muff TONE control that allows to tame harshness in the highs with extreme Muff's TONE settings, adding more usable tonal options (e.g. very tight on the low end but not painfully bright/too splattery). There is also the TIGHT knob that cuts out bass up to around 260 Hz.
5. Adds 2x VOLTS knobs that is controlling buffered voltage regulator for each clipping stage individually in the middle of the circuit, ranging from very low, when the transistor is close to around ≈18V (≈max supply voltage), allowing you to have kind of a splattery gated fuzz, which also reduces highs and lows content.
6. Adds CLEAN signal blend–in knob, unaffected by any of the EQ stages. Should be very nice to have for a bass guitar.
7. Accepts from 9V to 18V power supply, higher voltages make pre–boost gain stages cleaner and more high–headroom gain while the clipping stages can be set to whatever voltage you prefer individually (see VOLTS knobs, ≈3V–≈18V). I recommend 18V supply. 9V may cause CLEAN blended signal to clip on strong transients.
8. Improves DC filtering and adds extra RF filters, making the circuit less noisy, more stable.
9. Make the pedal output buffered, so your VOLUME knob does not affect output impedance of the pedal, it's always low.

Pots wiring (instead of cutting traces on the PCB):

GAIN/SUSTAIN pot pins 1 and 2 are soldered onto PCB, as designed by the PCB. Pin 3 though is soldered to the PCB TONE pot PCB hole 2 (thus sending the signal from T1 straight to T4).

TONE pot pins 1 and 3 are soldered onto the PCB as designed by the PCB. Pin 2 of the pot though is connected to pin 3 of the VOLUME pot (bypassing the PCB). So that the VOLUME pot is going right after the TONE pot.

None of the VOLUME pot pins are connected to the main PCB. Pin 3 is connected to TONE pot pin 2, pin 1 is soldered to the pot back/case (which is supposed to be grounded by touching the metal of the shielded pedal case). And pin 2 is connected to the extension mod PCB first buffer input (R34).

In the schematic stock parts for Muff circuit are the TRIANGLE one
but you can use whatever Muff variant you prefer.
Main Muff schematic is based on Musikding.de Muff DIY kit layout.
https://www.musikding.de/Der–Muff–Triangle–Distortion–kit

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